Identifiability Before Integration: Prospective Evidence Qualification for Pointwise Four-Terminal MOS Capacitance Reconstruction

Xianzheng Wang

Abstract—Finite completion of an incomplete four-terminal capacitance matrix does not establish that the completion is identifiable, numerically eligible, or suitable for promotion to a global charge representation. This work evaluates an auditable two-stage qualification interface. An upstream stage evaluates the complete, predeclared multi-mesh/frequency TCAD bundle and supplies a Boolean eligibility flag; the downstream stage receives that flag, schema-checked metadata, observed entries, and a synthetic mask. Missing entries are solved from row- and column-sum identities only when the scaled constraint system has zero nullity; otherwise, the frozen evaluator requests one full AC matrix at the same source and bias. Pointwise conservation and identifiability are also shown not to imply cross-bias integrability. The method was frozen before a prospective Benchmark F containing 72 off-grid point-pattern instances from six new TCAD source-condition clusters. Sixty-six instances were numerically eligible. No unsafe authorization was observed among 462 derived, predeclared mandatory-refusal conformance instances, and all 43 identifiable eligible instances were completed without acquisition. The frozen evaluator additionally authorized 23 null-space cases after same-point refresh; a post-F conformance audit found that it inherited point eligibility instead of independently reapplying the refresh truth-eligibility requirement in the original specification, so this branch is reported separately rather than counted as fully specification-conformant confirmation. For the 43 direct cases, matrix-relative held-out-entry error had median 6.01×10^{-10}, 95th percentile 1.31×10^{-8}, and maximum 2.87×10^{-7}. Relative to refreshing all 66 eligible points, requested refresh-job and cumulative job-wall ratios were 0.348 and 0.358; including shared qualification cost, the counterfactual total-workflow ratio was 0.742. The result supports bounded qualification of pointwise capacitance evidence, not universal corruption detection or a validated global charge-LUT.

Index Terms—Capacitance matrix, compact modeling, identifiability, MOSFET, prospective validation, TCAD.

I. Introduction

Charge-based MOS models derive displacement currents from terminal charges and therefore preserve relationships that independently fitted capacitance curves may violate [1]–[5]. Lookup tables can accelerate model evaluation and design-space exploration [6], but incomplete numerical tables introduce a prior question: what model class is actually authorized by the available evidence? Generic low-rank matrix completion addresses a different prior [7]; it does not replace device-specific conservation, provenance, and numerical-reference qualification. A solver may return finite values even when the missing entries are inconsistent, non-identifiable, poorly conditioned, semantically mislabelled, or unsupported by their numerical source.

This distinction is important for four-terminal differential capacitance. At one bias, row and column conservation identities constrain missing entries. Across biases, however, a global memoryless charge representation requires additional compatibility of the differential field. Pointwise completion and global integration are consequently different scientific claims. Treating them as one step can convert a local algebraic result into an unsupported compact-model assertion. Prior charge-based device work has likewise emphasized that a charge representation requires an existence and compatibility test [8]; the present work makes that boundary executable at the evidence-qualification interface and tests the bounded rule prospectively.

The present work addresses this problem through a preserved sequence of falsification, correction, and prospective validation. Development Benchmarks A–E exposed unsafe acceptance, over-refusal, missing-validation semantics, and an inaccurate DC-overwrite pathway. Numerical qualification then retained ACExtract capacitance as a bounded pointwise reference while rejecting the available terminal-charge derivatives as a global charge oracle. The corrected pointwise-v2 method was frozen before Benchmark F; F alone is the primary prospective test. In the implemented interface, the upstream qualification computation may read the complete TCAD bundle, whereas the downstream completion rule receives a Boolean truth_eligible flag, masked entries, and metadata. Raw-content hashes are verified separately at package level; they are not fields of a formal certificate object.

The contributions are:

1) an executable evidence-qualification sequence that separates numerical eligibility, semantics and provenance, consistency, rank/nullity, completion, and post-completion audit;
2) a one-refresh acquisition rule, minimal within the frozen action space, that performs no new simulation for a unique completion and requests one full same-point AC matrix only when null space remains;
3) an explicit downstream-observability boundary that limits corruption-detection claims to the evidence actually exposed to a decision rule;
4) a literature-grounded non-promotion proposition, with an analytical counterexample, showing that pointwise conservation and identifiability do not imply global charge integrability; and
5) a frozen, off-grid Benchmark F with independent

numerical recomputation and a post-F specification–evaluator conformance audit that preserves implementation deviations rather than silently rewriting the freeze.

The validated object is deliberately narrow: a numerically eligible, quasi-static, pointwise four-terminal capacitance matrix in the tested TCAD domain, qualified upstream and then handled by the downstream rule. Measurement validity, arbitrary-geometry transfer, RF behavior, a global charge surface, and circuit-level dynamics are not claimed.

II. Pointwise Evidence Object

A. Four-Terminal Formulation

Let terminal order be (g, d, s, b) and let

$$\mathbf{V} = (V_g, V_d, V_s, V_b)^{\mathsf{T}}. \tag{1}$$

At a fixed operating point, the signed ACExtract small-signal convention is

$$\delta\mathbf{i} = (\mathbf{A} + j\omega\mathbf{C})\,\delta\mathbf{V}, \tag{2}$$

where $\mathbf{A}$ is the real small-signal coefficient and $\mathbf{C}$ is the coefficient multiplying $j\omega$ in the ACExtract terminal response [9]. For indices i, j in the stated terminal order, C_{ij} is the signed coefficient of $j\omega\,\delta V_j$ in δi_i. Thus $\mathbf{C}$ is an operationally extracted, signed small-signal coefficient matrix; it is not defined here as $\partial\mathbf{Q}/\partial\mathbf{V}$ and this definition does not assume that a globally integrable terminal-charge map exists. The matrix entries use this signed convention directly; no additional minus sign is applied to an off-diagonal entry. If a charge-based convention reports an off-diagonal transcapacitance as $-\partial Q_i/\partial V_j$, that quantity must be translated before comparison. For the terminal-current orientation used here, common-mode invariance and Kirchhoff current conservation give distinct identities

$$\mathbf{C1} = 0, \qquad \mathbf{1}^{\mathsf{T}}\mathbf{C} = 0. \tag{3}$$

No reciprocal-symmetry condition $C_{ij} = C_{ji}$ is imposed; intrinsic MOS capacitances need not be reciprocal [1]. For reduced coordinates

$$\mathbf{u} = (V_{GS}, V_{DS}, V_{BS})^{\mathsf{T}} = \mathbf{T}\mathbf{V}, \quad \mathbf{T} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \tag{4}$$

Fig. 1 shows the original TCAD structure and the frozen decision sequence. The right panel is explicitly a method schematic; it is not a simulated structure or measured result.

B. Two-Stage Qualification Interface

The qualification object is intentionally separated from the completion object. The upstream stage receives complete coarse-, medium-, and fine-mesh evidence together with medium-mesh 1-Hz and 100-kHz evidence and computes the frozen numerical policy. The implemented downstream interface receives a Boolean truth_eligible flag, declared metadata, observed entries, and a prospectively assigned synthetic missingness mask. Source/bias identifiers and raw-file digests remain separate manifest and metadata records; the implementation does not construct or pass a single certificate object containing them. The downstream rule does not recompute numerical eligibility from the masked matrix. Consequently, the results qualify audited reuse of incomplete derived evidence; they do not establish a standalone imputer that can infer eligibility from an arbitrary incomplete matrix.

C. Incomplete Evidence as a Linear System

For a missing set M, unknown entries are stacked in deterministic row-major order into $\mathbf{x}$. Moving observed terms in (3) to the right-hand side yields

$$\mathbf{A}_M\mathbf{x} = \mathbf{b}_M. \tag{5}$$

Let $\mathbf{c}_{\text{obs}}$ denote the vector of observed entries in the same deterministic order. Because every unknown has the same physical unit, variable scaling is the identity. Each nonzero equation row is normalized to unit Euclidean norm:

$$\widetilde{\mathbf{A}} = \mathbf{D}_r\mathbf{A}_M, \qquad \widetilde{\mathbf{b}} = \mathbf{D}_r\mathbf{b}_M. \tag{6}$$

The frozen relative consistency residual is

$$r^*_{\text{rel}} = \frac{\min_{\mathbf{x}} \|\widetilde{\mathbf{A}}\mathbf{x} - \widetilde{\mathbf{b}}\|_2}{\max(\|\widetilde{\mathbf{b}}\|_2, \|\mathbf{c}_{\text{obs}}\|_2, 10^{-24})}. \tag{7}$$

Numerical rank uses singular values larger than $10^{-10}\sigma_{\max}$, following standard singular-value rank diagnostics [10], [11]. A system is pointwise identifiable when

$$\operatorname{rank}(\widetilde{\mathbf{A}}) = |M|, \quad \text{equivalently} \quad \dim\mathcal{N}(\widetilde{\mathbf{A}}) = 0. \tag{8}$$

For the pure row/column-sum support model, the missing entries form edges of a bipartite graph between row and column constraints. If m edges touch n vertices in c connected components, the incidence specialization gives

$$\operatorname{rank}(\mathbf{A}_M) = n - c, \qquad \dim\mathcal{N}(\mathbf{A}_M) = m - n + c. \tag{9}$$

Thus a forest is unique and each independent cycle adds one degree of freedom [12]. This graph statement is explanatory. Acquired rows, numerical scaling, and any future constraints can alter the actual system, so (8), not graph appearance, controls the decision.

III. Frozen Pointwise-v2 Qualification

The method applies a lexicographic sequence. Terminal order, coordinate order, units, source reference, source cluster, bias identifier, evidence role, and the lexical form of the provenance field are audited first. Byte-level raw-file integrity is a separate package-level SHA-256 audit, not the metadata-schema check. A point is then tested against the separately frozen ACExtract numerical-eligibility policy. Eligible incomplete evidence proceeds to consistency and rank/nullity analysis.

If (5) is consistent and has zero nullity, the minimum-residual solution is inserted without changing any observed entry. If nullity is nonzero, the method requests one

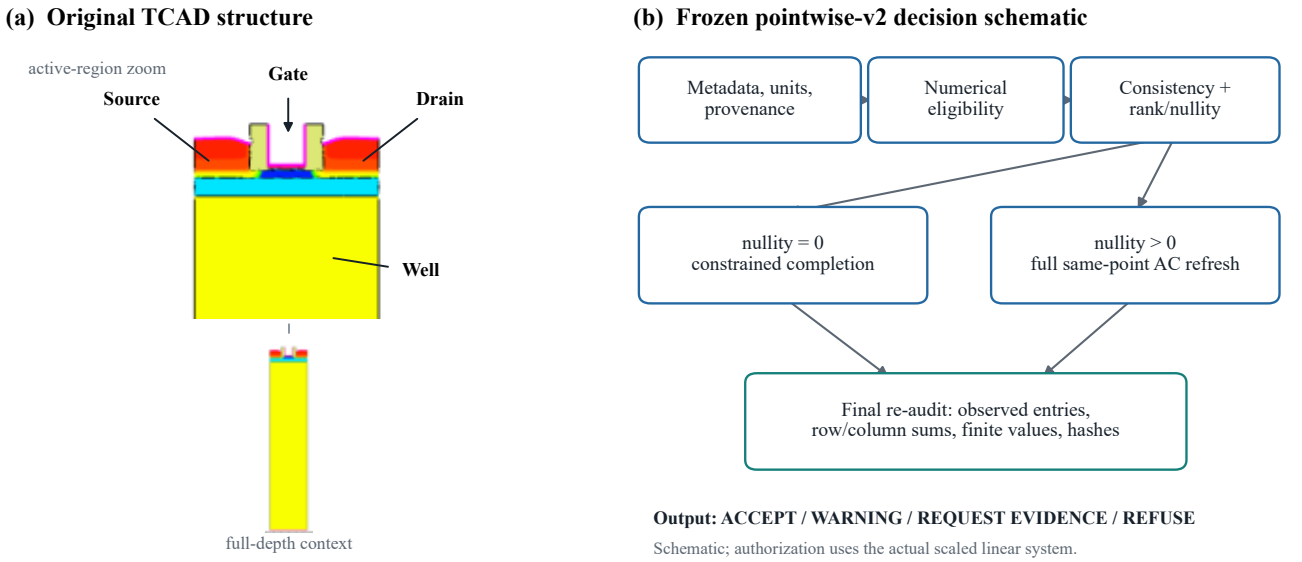


Fig. 1. (a) Original Sentaurus Visual export of the frozen process structure, shown as an active-region crop and full-depth context; terminal labels are editorial overlays and do not alter the simulated geometry. (b) Frozen pointwise-v2 decision schematic. Numerical authorization uses the actual scaled constraint system, not the drawing.

full medium-mesh 1-Hz ACExtract matrix at the same source and bias. The refresh is fit evidence, not independent validation evidence. The original pre-F specification requires the refresh itself to pass truth eligibility. The frozen Benchmark-F evaluator instead inherited eligibility from the same source/bias point and checked distinct raw-file digest, finite values, conservation consistency, and final conservation; it did not run a refresh-specific coarse/fine/frequency suite. This post-F conformance deviation is preserved. Direct completions are specification-conformant prospective results; refresh authorizations are evaluator-defined outcomes with post hoc agreement diagnostics, not evidence that the stricter refresh clause was prospectively satisfied.

Every finite output, including a zero-unknown or refreshed case, is re-audited. The frozen final residual is

$$r_C = \frac{\max\{\|\widehat{\mathbf{C}}\mathbf{1}\|_\infty, \|\mathbf{1}^\mathsf{T}\widehat{\mathbf{C}}\|_\infty\}}{\max(\|\widehat{\mathbf{C}}\|_F, 10^{-24})}, \tag{10}$$

and must not exceed 10^{-6}. Observed-entry preservation, finite values, terminal ordering, and metadata are rechecked; raw-content digests are handled by the separate package audit. A full-rank system with $\kappa_2 > 10$ receives a warning rather than silent acceptance.

Table I summarizes the frozen actions. No scientific threshold was changed after Benchmark F was opened.

For an authorized direct completion, the reported error is the matrix-relative held-out-entry Frobenius error

$$e_{\text{hold}} = \frac{\|\widehat{\mathbf{C}} - \mathbf{C}_{\text{medium}}\|_F}{\max(\|\mathbf{C}_{\text{medium}}\|_F, 10^{-24})}. \tag{11}$$

Observed entries are preserved, so the numerator is supported only on the prospectively masked entries, while the denominator is the norm of the complete medium-mesh reference. This is a held-out agreement metric within the same TCAD reference domain, not measurement error or physical validation. The acquisition analysis reports both the incremental refresh ratio and a counterfactual total-workflow ratio that includes the shared qualification suite.

IV. Two Information Boundaries

A. Observational Indistinguishability

Let $\mathcal{E}_{\text{down}}$ denote only the evidence actually exposed to the downstream rule: the eligibility flag, selected numerical entries, mask, and schema-checked metadata. It excludes the hidden complete matrix and package-level content digests. The following elementary observation fixes the scope of any downstream corruption-detection claim; it is not presented as a new mathematical theorem.

Proposition 1 (Evidence-operator boundary): Let $D = f \circ \mathcal{E}_{\text{down}}$ be any deterministic decision rule that reads only $\mathcal{E}_{\text{down}}$. If a nonzero perturbation $\Delta\mathbf{C}$ satisfies

$$\mathcal{E}_{\text{down}}(\mathbf{C} + \Delta\mathbf{C}) = \mathcal{E}_{\text{down}}(\mathbf{C}), \tag{12}$$

then $D(\mathbf{C} + \Delta\mathbf{C}) = D(\mathbf{C})$.

Proof: Equality of the evidence objects makes the arguments of f identical; deterministic evaluation therefore returns the same decision. ∎

The statement is intentionally bounded. It does not say that every algorithm fails, only that a rule restricted to the same downstream object cannot distinguish members of one equivalence class. The Benchmark-F internally row/column-conservative 1% perturbation is not an empirical instance of (12): the conformance evaluator was supplied with the perturbed complete matrix, so its numerical input changed. Instead, the 66/66 authorizations are a criterion-insufficiency stress test showing that the

TABLE I
Frozen Pointwise-v2 Decision Logic

Condition	Action
Invalid semantics, units, source, or provenance	Refuse
Numerically ineligible reference point	Refuse
Inconsistent conservation system	Refuse
Consistent, nullity > 0	Request Evidence
Unique, $\kappa_2 > 10$	Warning with finite matrix
Unique, stable, final audit passed	Accept
Requested full same-point refresh passes the frozen evaluator's implemented checks	Accept After Acquisition*

*The original specification also required refresh-specific truth eligibility. Benchmark F inherited point eligibility instead; the resulting conformance limitation is reported explicitly.

implemented conservation and metadata checks do not detect this particular conservative perturbation. Additional independent evidence could refine the downstream object and provide a different decision boundary.

B. Pointwise Qualification Does Not Imply Integrability

To analyze the separate promotion hypothesis, suppose that a differentiable, memoryless terminal-charge map $\mathbf{Q}(\mathbf{u})$ exists. Define

$$\mathbf{J}_u(\mathbf{u}) = \frac{\partial \mathbf{Q}}{\partial \mathbf{u}}, \qquad \mathbf{C}(\mathbf{u}) = \mathbf{J}_u(\mathbf{u})\mathbf{T}. \tag{13}$$

This relation is conditional: it connects the operational ACExtract coefficient field to a charge Jacobian only under the promotion hypothesis. On a simply connected bias domain, a continuously differentiable row of $\mathbf{J}_u$ can be a gradient of terminal charge only if its mixed partials are compatible [8], [13], [14]:

$$\frac{\partial (J_u)_{ij}}{\partial u_k} = \frac{\partial (J_u)_{ik}}{\partial u_j}. \tag{14}$$

Proposition 2 (Non-promotion boundary): Pointwise conservation and pointwise identifiability of $\mathbf{C}(\mathbf{u})$ do not imply the existence of a global memoryless terminal-charge map $\mathbf{Q}(\mathbf{u})$ satisfying (13).

Proof: Let $\mathbf{u} = (u_1, u_2, u_3)^{\mathsf{T}}$ and define the fully observed field

$$\mathbf{J}_u(\mathbf{u}) = \begin{bmatrix} 0 & u_1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -u_1 & 0 \end{bmatrix}. \tag{15}$$

Its terminal rows sum to zero, so $\mathbf{1}^{\mathsf{T}}\mathbf{J}_u = 0$. With $\mathbf{T}$ from (4), $\mathbf{C} = \mathbf{J}_u\mathbf{T}$ therefore satisfies both identities in (3): $\mathbf{T1} = 0$ gives $\mathbf{C1} = 0$, and $\mathbf{1}^{\mathsf{T}}\mathbf{J}_u = 0$ gives $\mathbf{1}^{\mathsf{T}}\mathbf{C} = 0$. Because the field is fully specified at each bias, pointwise missing-entry identifiability is trivial. However, for the gate row, $\partial (J_u)_{g2}/\partial u_1 = 1$ while $\partial (J_u)_{g1}/\partial u_2 = 0$, violating (14). Hence no global Q_g with this gradient exists, and no global four-terminal $\mathbf{Q}$ exists. ∎

This counterexample is analytical, not fitted to the TCAD data. It proves only the logical insufficiency of pointwise criteria for global promotion.

V. Prospective Benchmark F

A. Pre-Unblinding Freeze

Benchmark F was generated with Sentaurus Device X-2025.06 [9], using standard device-simulation practice [15], after pointwise-v2 code, thresholds, source list, off-grid biases, missingness assignment, corruption classes, success criteria, evaluator, and hashes had been frozen. It contains six previously unused temperature/work-function source-condition clusters in one simulator/model stack, not six independent fabricated devices. The off-grid coordinates are $V_{GS} \in \{0.15, 0.45, 0.75\}$ V, $V_{DS} \in \{0.20, 0.65\}$ V, and $V_{BS} \in \{-0.24, 0.13\}$ V, giving 72 derived point-pattern instances.

Each point has coarse-, medium-, and fine-mesh 1-Hz ACExtract evidence plus a medium-mesh 100-kHz result. The frozen F policy requires medium/fine relative difference at most 0.01, three-mesh deviation at most 0.02, 1-Hz/100-kHz difference at most 0.01, and normalized row/column residuals at most 10^{-6}. The numerical threshold value was reused from prior development, but the exact 1-Hz/100-kHz span is the F protocol and should not be described as an unchanged Stage-14 frequency experiment. Ineligible points remain in the attempted-point and provenance record.

Eighteen missingness patterns are balanced by source using a frozen seed. Twelve pattern classes are structurally identifiable forests and six contain null space. Every source receives eight identifiable and four non-identifiable instances. The complete medium-mesh matrices were generated first, and the missing entries were then assigned prospectively by software; thus Benchmark F is a controlled synthetic-masking benchmark. All 72 possible refresh jobs were generated and executed before unblinding, allowing an actual refresh-all cost baseline; the method revealed only the frozen requested subset for the counterfactual analysis. The 23-request figure is therefore an incremental acquisition result, not runtime saved during the executed benchmark. Operationally, this cost comparison represents downstream recovery after an incomplete derived-evidence interface has been received while the complete upstream matrix is unavailable or withheld; if the complete matrix remains directly accessible, retrieval rather than re-simulation is the appropriate baseline.

At each eligible point, seven mandatory-refusal instances alter terminal order, coordinate order, units, source reference, provenance-hash syntax, complete conservation, or incomplete-system consistency. The invalid-hash class tests metadata-schema rejection; byte-level integrity is established separately by the package SHA-256 audit. A separate exactly row/column-conservative 1% corruption is the explicit noncriterion boundary. Table II gives the predeclared criteria and observed values.

B. Prospective Results

Fig. 2 reports source eligibility, evaluator disposition, direct-case error, and acquisition cost in one view. Five source clusters retained all 12 points; source S5 retained 6/12, leaving 66 eligible instances. Of these, 43 were uniquely identifiable and accepted without acquisition. The other 23 eligible cases had null space and were authorized by the frozen evaluator after same-point refresh. The six ineligible points were not forced through the method. The first result is the fully specification-conformant prospective completion result; the second is retained as an evaluator-defined branch because refresh-specific truth eligibility was not independently recomputed.

For the 43 direct completions, matrix-relative held-out-entry error median, 95th percentile, and maximum were 6.0128×10^{-10}, 1.3123×10^{-8}, and 2.8673×10^{-7}, respectively. The maximum accepted final conservation residual across all frozen-evaluator outputs was 2.8815×10^{-7}. No graph/rank disagreement occurred. For the 23 refresh cases, post hoc same-TCAD primary/refresh NRMSE had median 4.89×10^{-12}, 95th percentile 1.23×10^{-9}, and maximum 3.68×10^{-9}; these are descriptive repeatability diagnostics, not predeclared refresh-eligibility criteria.

The protocol defines the counterfactual as refreshing every eligible point. The 23 requested refreshes therefore represent 23/66, or 34.85%, of that baseline. Their cumulative job wall time was 3728 s versus 10422 s for the 66 eligible refresh jobs, a ratio of 0.3577. Including the shared qualification suite (15568 s), the protocol-relative total-workflow ratio is $(15568+3728)/(15568+10422) = 0.7424$, corresponding to a 25.76% reduction. This is a counterfactual repeat-refresh/recovery comparison conditional on the downstream incomplete-evidence interface; it is not a saving in the initial generation of the upstream qualification bundle. The frozen evaluator had instead divided by all 72 executed refresh jobs, including six ineligible points, and recorded 0.3194, 0.3303, and 0.7185; those values are preserved only as historical evaluator outputs. All ratios are sums of per-job wall seconds, not elapsed parallel batch wall-clock, and no universal TCAD-project saving is claimed.

C. Refusal Performance and the Declared Boundary

All seven mandatory-refusal classes occurred once at each of 66 eligible points, totaling 462 derived conformance instances. None was authorized. Fig. 3 separately shows the conservative perturbation: all 66 were authorized because the perturbation preserved the checks used by the frozen criterion, although it changed the complete numerical matrix supplied to that stress-test evaluator. Reporting the two groups separately prevents a categorical conformance count from being mistaken for universal corruption detection.

D. Independent Recalculation

The frozen evaluator produced FULL_HYPOTHESIS_SUPPORTED. A second post-F checker did not import that evaluator; it rehashed all 144 raw outputs and reproduced its numerical aggregates and 11 recorded checks. A separate specification–evaluator–claim audit then identified two governance deviations: refresh truth eligibility was inherited rather than independently evaluated, and the cost denominator included six ineligible refreshes despite the protocol's eligible-point baseline. The original specification was restored to its pre-F SHA-256 value, the deviations were not retroactively edited away, and manuscript claims use the corrected protocol denominator. These checks provide computational recomputation, not physical independence from the shared TCAD model stack.

VI. Development Falsification and Promotion Boundary

Benchmarks A–E remain immutable development history and are not pooled with F as prospective evidence. Table III records why the final method differs from its predecessors.

The strongest surviving result is pointwise capacitance qualification. Promotion to a global memoryless $\mathbf{Q}(\mathbf{u})$ was separately tested and not supported by the tested constructions. Stage 15's dense AC-field construction had global relative residual 0.07840 versus an analytical discretization floor of 0.005245. Stage 16 reduced frequency to 1 Hz, but the 1-Hz and 100-kHz fields differed by only 1.57×10^{-12} while the 1-Hz global residual remained 0.08296 versus a same-grid floor of 0.01891. Stage 17 tested five explicit charge partitions; its best retained construction gave 0.16808 versus a 0.003861 floor. These frequency results are quasi-static consistency checks within the tested ACExtract configuration, not RF or dynamic validation.

The logical proposition in Section IV and the empirical results in Fig. 4 have different roles. Proposition 2 proves that pointwise criteria are insufficient in general. Stages 15–17 show only that the tested constructions did not overcome that gap for the available data. No universal nonexistence theorem is claimed.

VII. Discussion and Limitations

The result is positive in a bounded but useful sense. Within the tested two-stage interface, the downstream method distinguishes when conservation identities uniquely determine missing pointwise entries and when additional same-point evidence is requested. The

TABLE II
Predeclared Benchmark-F Criteria and Prospectively Observed Results

Criterion	Required	Observed	Result
Numerically eligible points	≥ 60 of 72	66 of 72	Pass
Sources with at least 8/12 eligible	≥ 5	5 of 6	Pass
Unsafe authorization, mandatory refusals	0	0 of 462	Pass
Identifiable no-acquisition coverage	≥ 0.95	$43/43 = 1.000$	Pass
All-valid post-acquisition coverage	≥ 0.95	$66/66 = 1.000$	Evaluator-defined outcome*
Identifiable held-out-entry error, 95th percentile	$\leq 5 \times 10^{-6}$	1.312×10^{-8}	Pass
Identifiable held-out-entry error, maximum	$\leq 10^{-5}$	2.867×10^{-7}	Pass
Maximum accepted conservation residual	$\leq 10^{-6}$	2.882×10^{-7}	Pass
Refresh-job ratio to eligible refresh all	≤ 0.5	$23/66 = 0.3485$	Pass
Cumulative refresh job-wall ratio	≤ 0.5	$3728/10422 = 0.3577$	Pass
Graph/numerical-rank disagreements	0	0	Pass

*Evaluator-defined outcome. The refresh branch did not independently satisfy the stricter refresh truth-eligibility clause in the original specification.

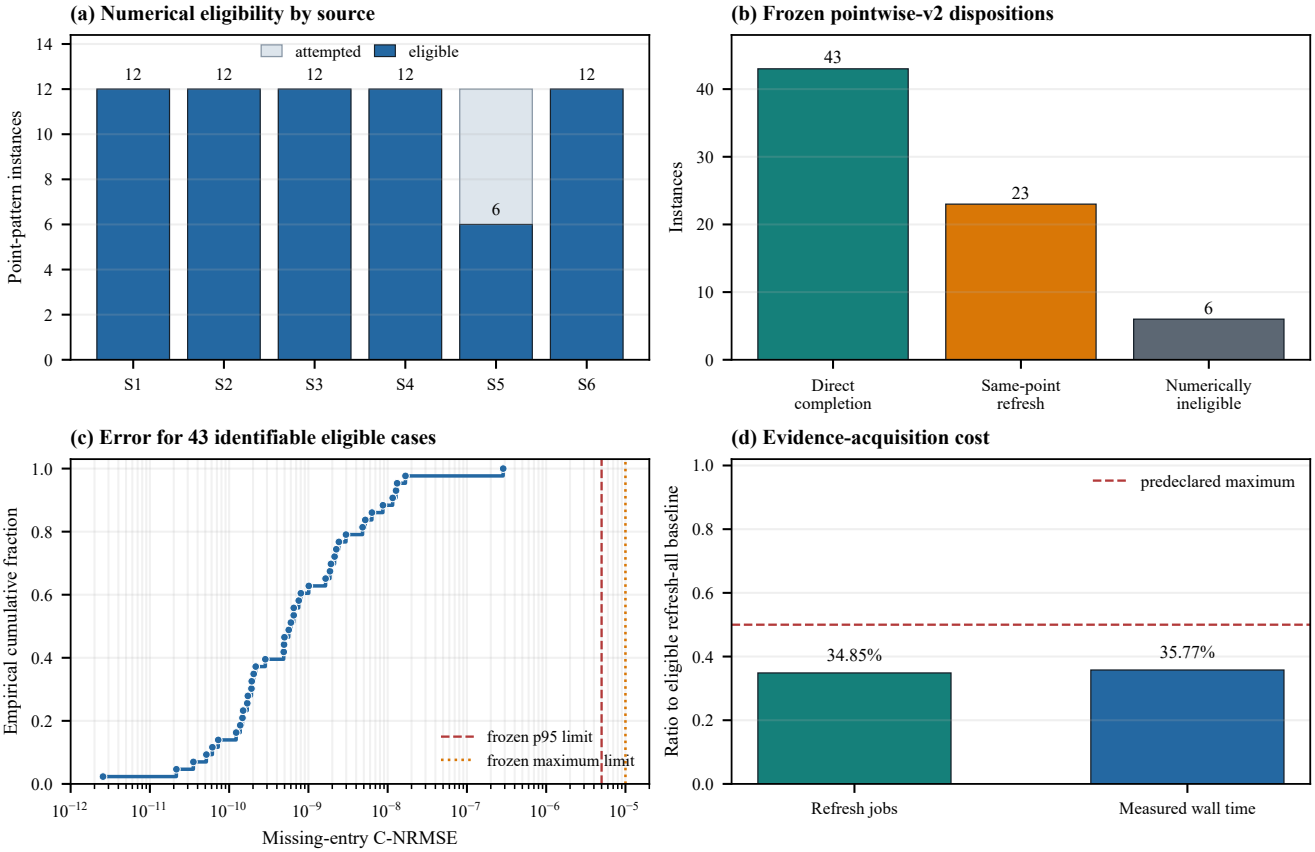


Fig. 2. Frozen Benchmark F with protocol-relative cost correction. (a) Attempted and numerically eligible derived instances by source-condition cluster. (b) Frozen-evaluator dispositions across all 72 instances. (c) Empirical cumulative distribution of the matrix-relative held-out-entry error for 43 specification-conformant direct cases; both predeclared error limits are shown. (d) Requested refresh jobs and cumulative job-wall time relative to refreshing all 66 eligible points.

specification-conformant direct branch completed 43/43 cases. The 23 refresh outcomes provide strong same-TCAD repeatability evidence but retain an explicit refresh-conformance limitation. Six points remain outside the numerical domain, and the conservative stress test shows that the declared criteria do not detect every internally conservative numerical perturbation.

The study has seven principal limitations. First, the upstream stage uses complete multi-mesh/frequency TCAD evidence to compute the eligibility flag; the downstream rule is therefore not a standalone incomplete-matrix eligibility detector. Second, the implementation passes a Boolean flag rather than a self-contained cryptographic certificate. Third, six controlled source-condition clusters are not a random sample of manufactured devices; the 72

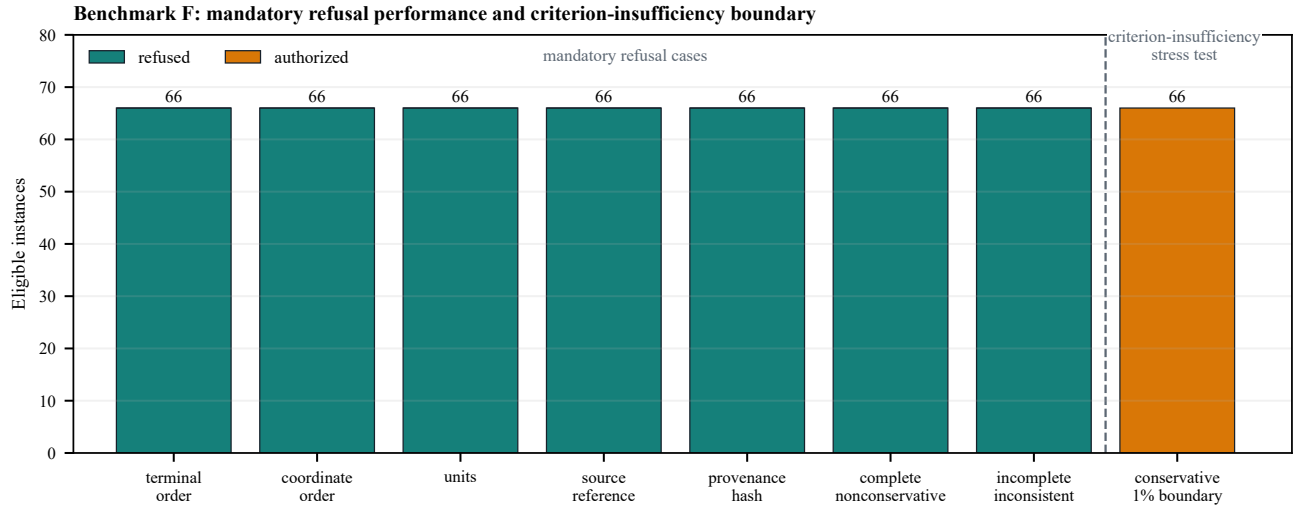


Fig. 3. Benchmark-F mandatory-refusal results and criterion-insufficiency boundary. Each class contains 66 eligible instances. The seven mandatory classes are all refused. The row/column-conservative 1% perturbations remain authorized because they preserve the checks used by the declared criterion; they are excluded from the safety-success count and are not treated as evidence-operator equality cases.

TABLE III
Compressed Development and Falsification History

Stage	Frozen observation	Consequence for the final method
A	Initial prototype separated decisions, reconstruction error, and circuit consequences.	Established that a finite output is not itself evidence sufficiency.
B	8/64 labelled unsafe instances were authorized.	Rejected the original strong fail-closed interpretation.
C	24/64 unsafe authorizations and 112/128 unnecessary refusals.	Rejected v2 as a usable authorization rule.
D	0/64 unsafe authorizations, but 0/192 finite outputs.	Rejected safety by universal refusal.
E	0/162 hard-error authorization and 241/432 categorical recovery, but later C-NRMSE p95 0.2624; the DC terminal-charge overwrite was ineligible.	Retired DC overwrite, empty-overlap scoring, and post-fill conservation bypass; E became development evidence only.

masks are derived instances, not 72 independent devices. Fourth, all evidence comes from one Sentaurus version and model stack. Fifth, the same-point refresh is fit evidence and its strict original refresh-eligibility clause was not independently implemented in F. Sixth, the hard-refusal result covers only the seven predeclared derived conformance classes. Seventh, no measurement, second simulator, PMOS, RF, arbitrary geometry, global charge surface, transient compact model, or circuit benchmark is validated.

These limitations do not negate the central methodological result. They define what further evidence would be required for promotion: an independent reference to distinguish evidence-equivalent objects, cross-bias compatibility for a global $\mathbf{Q}$, and dynamic validation only after the global representation passes. The method is therefore an auditable pre-construction qualification interface, not a substitute for compact-model validation.

VIII. Conclusion

An incomplete four-terminal capacitance matrix should be qualified before it is completed, and a completed pointwise matrix should be qualified again before it is integrated. The tested pointwise-v2 workflow applies an upstream eligibility flag, metadata-schema checks, conservation consistency, rank/nullity, constrained completion, null-space-triggered same-point refresh, and final re-audit. In prospective Benchmark F, 66 of 72 derived instances were eligible; all 43 identifiable cases were completed without acquisition, and no unsafe authorization was observed among 462 derived mandatory-refusal conformance instances. The frozen evaluator also authorized 23 refresh cases, but their original refresh-specific truth-eligibility requirement was not independently implemented and is therefore disclosed rather than silently treated as full specification conformance. For direct completions, matrix-relative held-out-entry error had median 6.01×10^{-10}, 95th percentile 1.31×10^{-8}, and maximum 2.87×10^{-7}. Relative to all 66 eligible refreshes, requested-job and

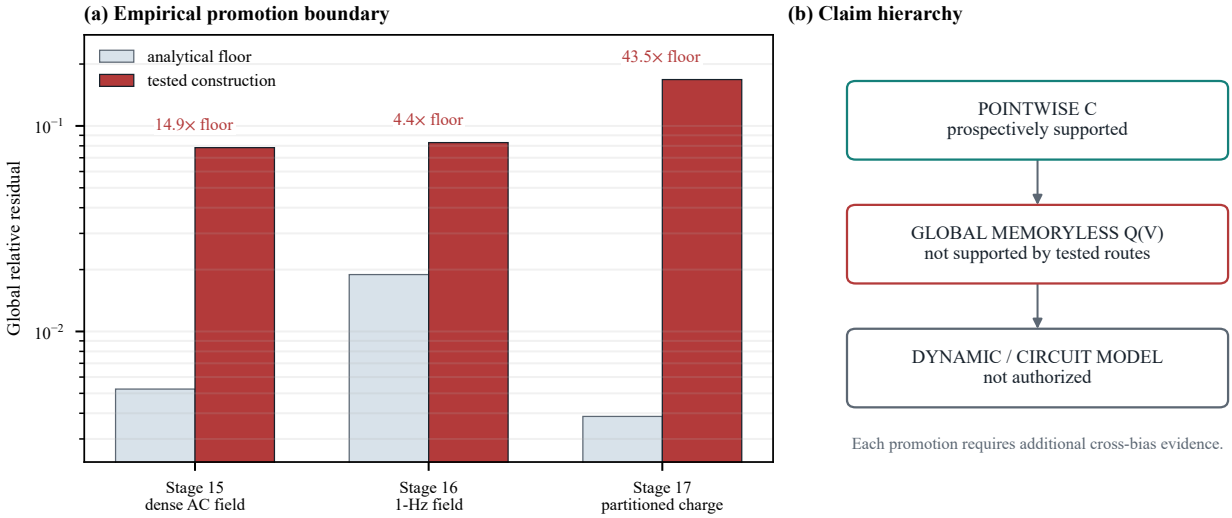


Fig. 4. Empirical promotion boundary. (a) Tested global constructions and their matched analytical discretization floors. These values show that the tested routes did not support a global memoryless charge representation; they do not prove that every possible global representation is impossible. (b) Frozen claim hierarchy. Pointwise success does not authorize a global charge or dynamic circuit model without additional cross-bias evidence.

cumulative job-wall ratios were 0.348 and 0.358; including qualification, the total-workflow ratio was 0.742. The conservative perturbation stress cases demonstrate a criterion boundary rather than universal corruption detection, and pointwise success did not support a global charge representation. The contribution is an auditable, bounded qualification result with preserved conformance limits, not universal safe reconstruction or a completed charge-LUT.

References

[1] D. E. Ward and R. W. Dutton, "A charge-oriented model for MOS transistor capacitances," IEEE Journal of Solid-State Circuits, vol. 13, no. 5, pp. 703–708, Oct. 1978.
[2] J. R. Brews, "A charge-sheet model of the MOSFET," Solid-State Electronics, vol. 21, no. 2, pp. 345–355, Feb. 1978.
[3] Y. Tsividis and C. McAndrew, Operation and Modeling of the MOS Transistor, 3rd ed. New York, NY, USA: Oxford University Press, 2011.
[4] C. C. Enz and E. A. Vittoz, Charge-Based MOS Transistor Modeling: The EKV Model for Low-Power and RF IC Design. Chichester, U.K.: Wiley, 2006.
[5] G. Gildenblat et al., "PSP: An advanced surface-potential-based MOSFET model for circuit simulation," IEEE Transactions on Electron Devices, vol. 53, no. 9, pp. 1979–1993, Sep. 2006.
[6] A. A. Youssef, B. Murmann, and H. Omran, "Analog IC design using precomputed lookup tables: Challenges and solutions," IEEE Access, vol. 8, pp. 134 640–134 652, 2020.
[7] E. J. Candès and B. Recht, "Exact matrix completion via convex optimization," Foundations of Computational Mathematics, vol. 9, pp. 717–772, 2009.
[8] M. Wiatr, P. Seegebrecht, and H. Peters, "Charge based modeling of the inner fringing capacitance of SOI-MOSFETs," Solid-State Electronics, vol. 45, no. 4, pp. 585–592, Apr. 2001.
[9] Synopsys, Sentaurus Device User Guide, Synopsys, 2025, version X-2025.06.
[10] G. H. Golub and C. F. V. Loan, Matrix Computations, 4th ed. Baltimore, MD, USA: Johns Hopkins University Press, 2013.
[11] L. N. Trefethen and I. D. Bau, Numerical Linear Algebra. Philadelphia, PA, USA: SIAM, 1997.
[12] R. Diestel, Graph Theory, 5th ed. Berlin, Germany: Springer, 2017.
[13] M. Spivak, Calculus on Manifolds. New York, NY, USA: W. A. Benjamin, 1965.
[14] J. M. Lee, Introduction to Smooth Manifolds, 2nd ed. New York, NY, USA: Springer, 2013.
[15] S. Selberherr, Analysis and Simulation of Semiconductor Devices. Vienna, Austria: Springer, 1984.